community due to its ease of use and powerful capabilities.

Docker Inc. was founded in 2013, and by 2014, Docker had become one of the fastest-growing open source projects in history. Major technology companies like Google, Microsoft, and Amazon quickly adopted Docker and began offering container-based services. The introduction of Docker Hub, a cloud-based registry for sharing container images, further accelerated adoption by creating a vast ecosystem of pre-built application containers.

Today, Docker has fundamentally changed how applications are developed, deployed, and managed, becoming an essential tool in modern DevOps practices and microservices architectures.

Virtual machines and their history

Virtual machines (VMs) represent a mature virtualisation technology that creates complete, isolated computing environments by emulating entire computer systems, including hardware components. Each VM runs its own operating system (guest OS) on top of a host operating system, with a hypervisor managing resource allocation and isolation between multiple VMs.

VMs provide complete isolation by virtualising all hardware components including CPU, memory, storage, and network interfaces. This comprehensive virtualisation allows multiple operating systems to run simultaneously on a single physical machine, each believing it has exclusive access to dedicated hardware resources.

The concept of virtualisation dates to the 1960s with IBM's CP/CMS system, which allowed multiple users to share mainframe resources efficiently. However, modern virtualisation, as we know it, began in the late 1990s with VMware's introduction of x86 virtualisation technology. VMware Workstation, released in 1999, brought virtualisation to desktop computers, followed by VMware ESX Server in 2001 for enterprise environments.

The 2000s saw rapid expansion in virtualisation adoption, driven by the need for better server utilisation and cost reduction. Microsoft entered the market with Hyper-V in 2008, while open source solutions like Xen and KVM provided alternatives to proprietary platforms. The rise of cloud computing in the 2010s further accelerated VM adoption, with providers like Amazon EC2, Google Compute Engine, and Microsoft Azure building their infrastructure primarily on virtualisation technology.

Virtual machines became the foundation of modern data centres, enabling server consolidation, disaster recovery, and flexible resource management that transformed enterprise IT infrastructure.

Scenarios where Docker is used

Docker excels in scenarios that prioritise speed, efficiency, and consistency across development and deployment pipelines. Microservices architecture represents one of Docker's most natural applications, where complex applications are broken down into smaller, independent services. Each microservice can be containerised separately, allowing teams to develop, deploy, and scale individual components independently while maintaining consistent runtime environments.

Continuous integration and continuous deployment (CI/CD) pipelines benefit enormously from Docker's consistency guarantees. Development teams can ensure that applications behave identically across development, testing, and production environments, eliminating the common "it works on my machine" problem. Docker containers can be built once and deployed anywhere, significantly reducing deployment-related issues.

Application modernisation projects frequently leverage Docker to containerise legacy applications without requiring extensive code rewrites. Organisations can package existing applications with their dependencies into containers, making them more portable and easier to manage while maintaining compatibility with modern orchestration platforms like Kubernetes.

Development environment standardisation becomes seamless with Docker, as entire development stacks can be defined in simple configuration files. New team members can spin up consistent development environments in minutes rather than spending days configuring local setups. This standardisation extends to testing environments, where identical conditions can be reproduced reliably.

Cloud-native applications and serverless architectures often rely on Docker containers due to their quick startup times and minimal resource overhead. Container orchestration platforms like Kubernetes, Docker Swarm, and cloud services like AWS ECS are built specifically around containerised workloads, making Docker the natural choice for modern cloud deployments.

Scenarios where VMs are used

Virtual machines remain the preferred choice for scenarios requiring complete isolation, diverse operating system support, or legacy application compatibility. Enterprise environments with strict compliance requirements often mandate VM-level isolation to ensure complete segregation between different applications or customer workloads. Financial institutions, healthcare organisations, and government agencies frequently rely on VMs to meet regulatory requirements that demand proven isolation capabilities.

Legacy application support represents a crucial VM use case, particularly when dealing with applications that require specific operating system versions or have deep integration with OS-level services. Many enterprise applications were designed with assumptions about dedicated hardware and operating system access that make containerisation